FOOT & ANKLE · NEET-SS / FET / INI-SS

Flatfoot and cavus foot

Part A: Adult acquired flatfoot (progressive collapsing foot deformity)

1. Terminology and anatomy

- **Adult acquired flatfoot deformity (AAFD)**, renamed **progressive collapsing foot deformity (PCFD)** (2020 consensus)**: complex 3D deformity** with **hindfoot valgus, forefoot abduction (talonavicular uncoverage), forefoot varus (supination), medial column instability (collapse of the arch)**, and later **ankle valgus**.
- **Tibialis posterior (TP): strongest invertor; primary dynamic stabiliser of the medial longitudinal arch**; inserts mainly on the **navicular tuberosity** with slips to the cuneiforms, cuboid and metatarsal bases 2-4; **locks the transverse tarsal (Chopart) joint** (TN and CC axes become non-parallel: rigid lever for push-off); **antagonist: peroneus brevis**.
- **Hypovascular zone** of TP tendon: **behind and just distal to the medial malleolus** (degeneration site).
- **Static stabilisers: spring ligament (calcaneonavicular: superomedial part most important)**, **deltoid ligament (deep)**, **plantar fascia (windlass mechanism)**, **talocalcaneal interosseous ligament**.
- **Posterior tibial tendon dysfunction (PTTD)** was the classic cause, but **spring ligament failure** is now recognised as central; TP failure is often secondary.

2. Causes

- **TP tendon degeneration (commonest in middle-aged obese women; hypertension, diabetes, steroid injections, seronegative arthropathy)**.
- **Others: inflammatory arthritis (RA)**, **trauma (TP laceration/rupture, Lisfranc injury)**, **neuropathic (Charcot)**, **tarsal coalition (adult presentation)**, **accessory navicular**, **degenerative TMT/naviculocuneiform arthritis**, **iatrogenic**.

3. Clinical evaluation

- **Medial ankle pain along the TP tendon** (early), **lateral sinus tarsi/subfibular impingement pain** (late: calcaneofibular impingement).
- **"Too many toes" sign** (**forefoot abduction seen from behind**).
- **Single-limb heel rise test**: **inability to perform / no heel inversion on heel rise** (**most sensitive test for TP dysfunction**); **double heel rise** to check flexibility (hindfoot inverts if flexible).
- **Flexibility of the hindfoot (subtalar)** and **forefoot (forefoot varus/supination: compensatory fixed or flexible)**: **assess forefoot position with the hindfoot held in neutral**.
- **Gastrocnemius/Achilles contracture** (**Silfverskiöld test**: almost universal).
- **Medial column instability** (first TMT/naviculocuneiform plantar sag), **deltoid laxity (ankle valgus)**.

4. Imaging

- **Weight-bearing radiographs (AP and lateral foot, AP ankle, hindfoot alignment view (Saltzman))**:
  - **Lateral talo-first metatarsal angle (Meary angle): normal 0 degrees (collinear)**; **over 4 degrees convex plantar = flatfoot** (**mild 4-15, moderate 15-30, severe over 30**),

  - **Calcaneal pitch angle (normal about 20-30 degrees; reduced in flatfoot)**,
  - **AP talonavicular coverage angle (over 7 degrees) / talonavicular uncoverage percentage (over 40%)**: forefoot abduction,
  - **AP talo-first metatarsal angle** (**abduction**),
  - **Medial cuneiform-floor height** (reduced), **sinus tarsi impingement**, **subfibular impingement**.
  - **Ankle: talar tilt (valgus) in the mortise** (deltoid failure).
- **Weight-bearing CT** (**3D, subfibular impingement, middle facet subluxation**: **peritalar subluxation**).
- **MRI** (TP tendon tears/tendinosis, spring ligament), **ultrasound**.

5. Classification

Johnson and Strom (modified by Myerson/Bluman)

Stage	TP tendon	Hindfoot	Treatment
I	**Tenosynovitis/tendinosis; normal length**	**Normal alignment; can single heel rise** (painful)	**Non-operative (immobilisation, NSAIDs, orthoses)**; **tenosynovectomy**

Stage	TP tendon	Hindfoot	Treatment
II (A/B/C)	**Elongated/dysfunctional**	**Flexible flatfoot** (**IIA mild abduction; IIB abduction over 30-40% TN uncoverage; IIC fixed forefoot varus**)	**Joint-sparing reconstruction**: **FDL transfer to the navicular + medializing calcaneal osteotomy (MDCO) ± lateral column lengthening (Evans) ± Cotton osteotomy ± gastrocnemius recession ± spring ligament repair**
III	**Ruptured/degenerated**	**Rigid flatfoot (fixed hindfoot valgus)**	**Triple arthrodesis** (or **double/modified: TN + subtalar**)
IV (A/B)		**+ ankle valgus (deltoid failure)**: **IVA flexible (correctable), IVB rigid (tibiotalar arthritis)**	**IVA: deltoid reconstruction + hindfoot correction**; **IVB: tibiotalocalcaneal (TTC) or pantalar arthrodesis, or TAR + hindfoot correction**

PCFD consensus classification (2020)

- **Stage 1 (flexible) or 2 (rigid)**, with **classes A (hindfoot valgus), B (midfoot/forefoot abduction), C (forefoot varus/medial column instability), D (peritalar subluxation/dislocation), E (ankle instability: valgus)**; each scored as flexible or rigid.

6. Treatment details

- **Non-operative (all stages first, mainstay in elderly/low demand): CAM boot or short leg cast (acute tenosynovitis, 4-8 weeks)**, **orthoses (medial arch support + medial heel wedge; UCBL)**, **ankle-foot orthosis (Arizona brace)** for later stages, **physiotherapy (eccentric TP strengthening, Achilles stretching)**, weight loss. **Avoid steroid injection into the TP tendon sheath (rupture)**.
- **Flexible deformity (stage II) procedures:**
  - **FDL transfer to the navicular** (**FDL is in phase with and adjacent to the TP; weaker (about 30%) but adequate when combined with bony correction**; **FHL** alternative; **interconnection (Henry knot) allows the FHL to flex lesser toes**),
  - **Medializing displacement calcaneal osteotomy (MDCO)** (**about 10 mm medial shift**: **converts the Achilles tendon pull into an invertor (medial to the subtalar axis)**; **protects the tendon transfer**; **sural nerve and lateral calcaneal branch at risk; the tibial nerve/tarsal tunnel at risk with large shifts**),
  - **Lateral column lengthening (Evans osteotomy: anterior calcaneus between the anterior and middle facets, about 10-15 mm from the CC joint; or CC distraction arthrodesis)** for **forefoot abduction (TN uncoverage over about 40%)**: **lateral column overload (lateral foot pain: commonest complication)**, **CC arthritis**, **sural nerve injury**, non-union,
  - **Cotton osteotomy (plantarflexion opening-wedge osteotomy of the medial cuneiform)** or **first TMT arthrodesis (Lapidus)** for **residual forefoot varus/medial column instability**,
  - **Spring ligament repair/reconstruction**,
  - **Gastrocnemius recession (Strayer) / Achilles lengthening (Hoke)** for equinus,

  - **Subtalar arthroereisis** (sinus tarsi implant: limited role in adults; mainly paediatric).
- **Rigid deformity (stage III): triple arthrodesis (subtalar, TN, CC)** through **medial and lateral incisions (or single medial incision: protects lateral skin)**; **TN joint is the key joint (fusion of TN alone eliminates almost all hindfoot motion)**; **position: hindfoot about 5 degrees valgus, neutral forefoot rotation (plantigrade), neutral abduction**; **complications: non-union (TN commonest site), malunion, adjacent joint arthritis (ankle, naviculocuneiform)**.
- **Stage IV: deltoid ligament reconstruction** (flexible) or **TTC/pantalar arthrodesis** (rigid with arthritis); **total ankle replacement with hindfoot realignment**.

Part B: Pes cavus

7. Definition and causes

- **Elevated medial longitudinal arch (high arch), often with hindfoot varus**; **forefoot-driven (plantarflexed first ray) or hindfoot-driven**.
- **About two-thirds have an underlying neurological cause**: **Charcot-Marie-Tooth disease (HMSN type I: commonest; autosomal dominant; PMP22 duplication on chromosome 17p)**, **spinal cord pathology (tethered cord, diastematomyelia, syringomyelia, tumour, spinal dysraphism)**, **cerebral palsy**, **poliomyelitis (India: residual polio)**, **Friedreich ataxia**, **stroke**, **muscular dystrophy**. **Also: idiopathic/familial (subtle cavovarus)**, **residual clubfoot**, **trauma (compartment syndrome: Volkmann's ischaemic contracture: claw toes; malunited talar neck fracture)**.
- **Unilateral cavus = spinal cord lesion until proved otherwise: MRI whole spine. Bilateral cavus: think CMT (nerve conduction studies, genetic testing)**.

8. CMT pathomechanics

- **Muscle imbalance (pairs of agonist/antagonist; weakest muscles first: tibialis anterior and peroneus brevis; intrinsics):**
  - **Weak tibialis anterior vs strong peroneus longus**: **plantarflexed first ray (forefoot pronation/valgus)**,
  - **Weak peroneus brevis vs strong tibialis posterior**: **hindfoot varus, forefoot adduction**,
  - **Intrinsic weakness**: **claw toes** (**EDL recruited to dorsiflex the ankle: hyperextends MTPJs**),
  - **Plantar fascia contracture** (windlass: elevates the arch).
- **Tripod effect**: **plantarflexed first ray forces the hindfoot into varus on weight bearing (forefoot-driven hindfoot varus)**.
- **Lateral ankle instability, peroneal tendon tears, fifth metatarsal stress fractures, lateral foot callosities, stress fractures, ankle arthritis (varus)**.

9. Evaluation

- **Peek-a-boo heel sign** (**medial heel pad visible from the front with the feet parallel: hindfoot varus**).
- **Coleman block test** (**block under the lateral foot (heel and lateral forefoot), allowing the first ray to drop off the edge**): **hindfoot varus corrects = flexible hindfoot, deformity is forefoot-driven** (treat the forefoot: **first metatarsal dorsiflexion osteotomy + plantar fascia release**); **does not correct = fixed hindfoot varus (needs calcaneal osteotomy or arthrodesis)**.
- **Muscle power testing** (MRC grading of each muscle: guides tendon transfers), **neurological examination** (reflexes, sensation, spine), **calluses (under first and fifth metatarsal heads)**.
- **Radiographs: Meary angle (talo-first metatarsal) convex dorsally (negative: over -4 degrees = cavus)**, **calcaneal pitch angle increased (over 30 degrees: calcaneocavus)**, **Hibbs angle**

(calcaneus axis vs first metatarsal axis: **under 150 degrees** = cavus), **"double talar dome"** (ankle AP/lateral: hindfoot varus/external rotation), **fibula posterior**.

10. Treatment

- **Non-operative: orthoses (lateral forefoot post + recess under the first metatarsal head)**, **lateral heel wedge** (for flexible hindfoot), **AFO** (foot drop), **physiotherapy**, **footwear**.
- **Operative (individualised; "flexible deformities need soft-tissue/tendon procedures, fixed deformities need bony procedures, arthrodesis for severe rigid deformity or arthritis"):**
  - **Soft tissue: plantar fascia release (Steindler)**, **gastrocnemius recession/Achilles lengthening** (**in calcaneocavus (polio/spina bifida) the Achilles is weak: do not lengthen**).
  - **Tendon transfers (balance forces): peroneus longus to peroneus brevis transfer** (**removes plantarflexion of the first ray; augments eversion**), **tibialis posterior transfer to the dorsum (through the interosseous membrane to the lateral cuneiform)** for **foot drop**, **Jones transfer (EHL to the first metatarsal neck + hallux IPJ fusion)** for **clawed hallux**, **Hibbs transfer (EDL to the lateral cuneiform/cuboid)** for **claw lesser toes**, **flexor-to-extensor transfer (Girdlestone-Taylor)** for flexible claw toes.
  - **Bony (joint-sparing): dorsiflexion closing-wedge osteotomy of the first metatarsal base** (**forefoot-driven cavus**), **lateral closing-wedge calcaneal osteotomy (Dwyer)** or **lateralising calcaneal (Z) osteotomy** (**fixed hindfoot varus**), **midfoot osteotomy (Cole: dorsal closing wedge through the navicular-cuneiform/cuboid; Japas V-osteotomy; Akron dome)** for **midfoot cavus**, **anterior tibial osteotomy** rarely.
  - **Arthrodesis: triple arthrodesis** (**rigid deformity, arthritis, salvage**; **worse long-term results in CMT (ankle/midfoot**

arthritis): **avoid when possible in young patients**).
    - **Lateral ankle ligament reconstruction** (Broström) **must be combined with correction of the varus** (otherwise fails).

11. Paediatric flatfoot (brief link)

- **Flexible flatfoot: physiological; no treatment unless painful + tight heel cord (calcaneal lengthening + gastrocnemius recession)**.
- **Rigid flatfoot: tarsal coalition, CVT, accessory navicular, inflammatory, neuromuscular**: see paediatric foot notes.

12. Other conditions

- **Tibialis anterior tendon rupture: elderly men (sixth-seventh decade)**, **degenerative**, **foot slap/steppage gait**, **palpable gap/mass at the anterior ankle**, **EHL/EDL substitution (toe clawing on dorsiflexion)**; **repair/reconstruction (EHL transfer, graft) in active patients**; **AFO in low-demand**.
- **Accessory navicular (os naviculare)** in adults: **Kidner procedure (excision + TP advancement)**.
- **Tarsal tunnel syndrome: tibial nerve compression beneath the flexor retinaculum (laciniate ligament)**; **space-occupying lesions (ganglion, varicosities, lipoma, accessory muscles: flexor digitorum accessorius longus)**, **hindfoot valgus (traction)**, trauma; **burning plantar paraesthesia (sparing the heel (calcaneal branch arises proximal) in some cases), Tinel sign, dorsiflexion-eversion test**; **NCS**; **MRI**; **decompression (release of the flexor retinaculum + abductor hallucis fascia (medial and lateral plantar nerve tunnels))**: **results less predictable than CTS (about 50-75%)**, best with a **space-occupying lesion**.
- **Baxter nerve (first branch of the lateral plantar nerve / inferior calcaneal nerve) entrapment**: **chronic heel pain (medial)**

mimicking plantar fasciitis; **abductor digiti minimi atrophy on MRI**; decompression with plantar fascia release.

- **Plantar fasciitis (heel pain): degenerative fasciosis at the medial calcaneal tubercle origin**; **first-step morning pain (post-static dyskinesia)**; **calcaneal spur (in FDB origin) is incidental**; **non-operative (about 90%: plantar fascia-specific stretching (best), Achilles stretching, heel cups, night splints, NSAIDs)**; **steroid injection (risk of fat pad atrophy and plantar fascia rupture)**, **ESWT**; **surgery (partial plantar fascia release (medial third to half: avoid complete release: lateral column overload)** ± **Baxter nerve release, gastrocnemius recession)** after **6-12 months**.

Exam pearl: **PCFD/AAFD: TP + spring ligament failure; too many toes; can't single-heel-rise (most sensitive). Johnson-Strom: I tendinosis (conservative), II flexible (FDL transfer + MDCO ± Evans ± Cotton ± gastroc recession), III rigid (triple arthrodesis), IV ankle valgus (deltoid recon or TTC/TAR). MDCO converts Achilles to an invertor; Evans = lateral column overload.**

Exam pearl: **Meary angle: plantar convex = flatfoot, dorsal convex = cavus. Calcaneal pitch high = calcaneocavus. TN joint key to triple arthrodesis (TN non-union commonest); position 5 degrees valgus.**

Exam pearl: **Cavus: unilateral = spinal MRI; bilateral = CMT (PMP22). CMT: weak TA vs strong PL (plantarflexed first ray), weak PB vs strong TP (varus). Coleman block: corrects = forefoot-driven (first MT dorsiflexion osteotomy + plantar fascia release + PL to PB); does not correct = add Dwyer/lateralising calcaneal osteotomy. Jones transfer for clawed hallux. Broström fails without varus correction.**

> Exam pearl: **Tarsal tunnel: flexor retinaculum, space-occupying lesion gives best release results. Plantar fasciitis: stretching best; partial release only. Baxter nerve: ADM atrophy.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.